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AHP-entropy Weight Method Based Comprehensive Evaluation Approach for Distribution Network Research

Zhan Shen
China Southern Power Grid Company Limited
Guangzhou 510663, China
1226117175@qq.com

* Kecheng Wu
China Southern Power Grid Company Limited
Guangzhou 510663, China
* Corresponding author: zymatcha@126.com

Abstract—**With the advent of the integrated energy system, diverse energy sources can be converted and substituted for one another, leading to significant transformations in the power system's operation mode. In the realm of smart integrated energy planning and development, addressing the thorough assessment of intelligent distribution network planning has become a crucial concern. To tackle this challenge, this paper introduces an approach that integrates subjective and objective weighting methods for a thorough assessment. At the outset, a series of assessment criteria is devised and quantified to match the features of the intelligent integrated energy system. Following that, a holistic assessment model is created, utilizing an amalgamated approach of subjective and objective weighting methods by integrating the AHP-entropy weight method to assign weights to the criteria. Real-world data from the power grid is then evaluated and scored. The precision and efficacy of the proposed method are validated through the obtained scoring results.**

Keywords-integrated energy system; future distribution; intelligent integrated energy planning and development; an approach integrating subjective and objective weighting techniques

I. INTRODUCTION

The rapid development of the Internet industry has brought great changes to people's production and life, and the energy industry as a traditional industry has entered a new round of reform and innovation, in this context for the pursuit of multi-energy coupling interconnection and various forms of energy efficient, clean and safe use of the "energy Internet" came into being [1-3]. In the energy Internet, the most representative and implementation is the integrated energy system [4-6]. In the smart grid, the smart distribution network is particularly critical. Under the background of smart comprehensive energy construction, it is of great significance to evaluate the smart distribution network comprehensively [7-9]. The existing comprehensive evaluation has achieved a lot of results, but it is no longer suitable for the rapid development of the future smart comprehensive energy system of distribution network [10-12]. Creating a comprehensive evaluation index system and methodology for intelligent energy systems that aligns with the national context of China holds significant practical importance [13-16].

Currently, extensive research is dedicated to the comprehensive assessment field [17]. This encompasses two primary areas of focus: the scrutiny of the analysis and evaluation index system, and the application of evaluation methods based on existing index systems [18-21]. When delving into the analysis and evaluation of index systems, practical experience is often considered [22-24]. Despite its practicality and intuitiveness, this approach is susceptible to significant subjective influences. Another evaluation approach concentrates on the analysis of individual indices, such as economy, reliability, and environmental protection [25-28]. In accordance with the guidance in Reference [29], an index system for evaluation was established, encompassing five aspects: power supply capacity, reliability, quality, economy, and sufficiency. The analytic hierarchy process was utilized to construct an evaluation model. Reference [30] assesses the technology, economy, and environment facets of integrated energy systems using the analytic hierarchy process. Reference [31] is grounded in the construction of a target grid evaluation system, utilizing fuzzy mathematics for a quantitative assessment of distribution networks. Chen B et al. employed the inverse entropy method in network analysis to determine the weight and scoring function of indices, addressing the issue of cross and mutual influences arising from inaccurate weights [32]. References [33-34] construct a comprehensive evaluation index system that includes economic, social, and environmental aspects when evaluating comprehensive energy systems. In contrast to previous literature focusing on a single user group, Reference [35] proposed a variable weight calculation method based on shape similarity due to the traditional variable weight calculation method's challenges in selecting state variable weight vectors and weak operability. To mitigate the impact of subjective selection disparities and enhance the credibility of evaluation outcomes, Reference [36] introduced the combined fuzzy comprehensive evaluation method. Simultaneously, in Reference [37], an analytic hierarchy process and an enhanced entropy weight method were employed to determine weight allocations for diverse indices, leading to the establishment of a VIKOR multi-criteria evaluation system. Furthermore, Reference [38] devised a system equivalent model grounded in weighted directed graphs. It introduced a method for calculating discrete energy flow and evaluating energy efficiency. Reference [39] offers a comprehensive overview of evaluating integrated energy systems. In summary, a balanced exploration of the evaluation system should consider both subjective and objective perspectives. Therefore, it is necessary to study a subjective and objective evaluation method to conduct comprehensive evaluation, which will be more comprehensive and complete, and the evaluation results will be more accurate and convincing.

In conclusion, this study suggests a holistic weighting approach utilizing the AHP-entropy weight method for the prospective intelligent integrated energy system in distribution networks. Initially, an evaluation index system is formulated, with primary-level indices including economy, reliability, environmental protection, and interactivity. Subsequently, a comprehensive weighting method is employed to assign weights to these indices. Finally, actual data from a city's distribution network is utilized for scoring and calculating the indices. The outcomes of the numerical examples affirm the effectiveness of the proposed method, highlighting its potential to offer additional decision-making support for the development and enhancement of distribution networks.

II. EXAMPLE ANALYSIS

A. *Comprehensive evaluation index*

Establishing a thorough evaluation index system lays the groundwork for evaluating a smart grid at the regional level. This initiation involves taking into account the current status and objectives of China's smart grid, covering all aspects of smart grid functionalities. It emphasizes the planning advantages and technical characteristics unique to the regional smart grid [40]. The principal indices mainly include the following four components:

1) *Economic index*

Economic and technical indices can capture the interdependence among various technical and economic phenomena and processes. They provide insights into the technical proficiency, managerial competence, and economic outcomes of production and business activities.

- Expenditure Aspect: This covers the initial investment, operational and maintenance costs, and costs related to failures. The starting costs include expenses for design and planning, acquiring equipment, and construction and installation. It's noteworthy that design and planning, as well as equipment purchase costs, may exhibit notable regional variations, introducing regional distinctions through specific coefficients. Operational and maintenance costs primarily involve day-to-day operational expenses, maintenance expenditures, and other related costs influenced by the line loss rate. Failure costs encompass both power outage losses and losses linked to failures.

$$C_a = C_i + C_{OM} + C_F \tag{1}$$

In the formula: C_a refers to the investment cost, C_i refers to the initial investment cost, C_{OM} refers to the operation and maintenance cost, C_F refers to the failure cost.

- Comprehensive line loss rate: It refers to the electrical energy lost on the line as a percentage of the electrical energy out put in the first section of the line

$$\Delta P = \frac{Q_s - Q_p}{Q_s} \tag{2}$$

In the formula: ΔP refers to the comprehensive line loss rate, Q_s refers to the power supply quantity, Q_p refers to the electricity consumption.

- Additional load per unit investment: It pertains to the proportion of the added load in relation to the investment expenditure.

$$Q_l = \frac{Q_{\ln} - Q_{ll}}{Q_{al}} \tag{3}$$

In the formula: Q_l refers to the additional load per unit investment, $Q_{\ln}$ refers to the current year's load, Q_{ll} refers to the previous year's load, Q_{al} refers to the cost of power grid investment last year.

2) *Reliability index*

The power system's reliability is gauged through quantitative reliability indices, which include the probability, frequency, and duration of faults affecting power users. Additionally, it encompasses the anticipated power loss and expected energy loss resulting from faults.

- Total Pass Rate: This indicates the percentage of the number of lines that meet the N-1 criterion relative to the total number of lines.

$$K_{pr} = \frac{N_N}{N_{al}} \tag{4}$$

In the formula: K_{pr} refers to the total pass rate, N_N refers to the number of lines that satisfy N-1, N_{al} refers to the total number of lines.

- Transformer Overload Rate: This parameter relates to the proportion of main transformers operating under heavy load conditions compared to the total number of main transformers. Main transformers considered heavily loaded are those with a maximum load rate exceeding 80% and a single duration exceeding 2 hours.

$$K_{ol} = \frac{N_{ol}}{N_{mt}} \tag{5}$$

In the formula: K_{ol} refers to the overall overload rate, N_{ol} refers to the number of heavy duty main transformers, N_{mt} refers to the total number of major transformers.

- Average Power Outage Time of Users: This metric represents the proportion of the total power outage time for users in a given unit of time relative to the total number of users.

$$T_{ap} = \frac{T_i}{N_i} \tag{6}$$

- In the formula: T_{ap} refers to the customers' average outage time, T_i refers to the total customer outage time, N_i refers to the total number of customers.

3) Environmental protection index

In the face of the difficulties posed by global warming and the energy crisis, countries globally are actively promoting a green economy and the establishment of eco-friendly power networks. The idea behind building a sustainable power grid involves establishing an energy-efficient and eco-friendly electricity network, guided by the principles of "safety, efficiency, environmental sustainability, and harmony" [41].

- Penetration Rate of Distributed Power Supply: This indicates the proportion of the installed capacity of distributed power supply to the capacity of 110kV public substation.

$$K_p = \frac{Q_{dg}}{Q_r} \tag{7}$$

In the formula: K_p refers to the permeability of distributed generation; Q_{dg} refers to the installed capacity of distributed power supply; Q_r refers to the 110kV public substation capacity.

- Proportion of Renewable Energy Generation Capacity: This represents the share of the installed generation capacity from renewable sources concerning the overall power generation capacity of the area.

$$K_r = \frac{Q_{re}}{Q_{al}} \tag{8}$$

In the formula: K_r refers to the installed capacity ratio of renewable energy; Q_{re} refers to the installed capacity of renewable energy power generation, Q_{al} refers to the total installed capacity.

- Proportion of Renewable Energy in Total Generating Capacity: This indicates the percentage of renewable energy within the overall generating capacity.

$$K_p = \frac{Q_{rp}}{Q_{al}} \tag{9}$$

In the formula: K_p refers to the ratio of renewable energy generation to electricity generation, Q_{rp} refers to the renewable energy generation, Q_{al} refers to the total installed capacity.

4) Environmental protection index

Interactivity stands out as a crucial aspect of intelligent distribution networks. The interactivity of distribution network changes the single directionality of source and load of traditional distribution network, and allows the distribution power to be connected to the grid, changes the single power flow of traditional distribution network, and improves the flexibility of network frame.

- Rate of Power Distribution Information Retrieval: This indicates the percentage of distribution transformers configured for information collection in comparison to the overall number of distribution transformers.

$$K_{as} = \frac{N_{as}}{N_{um}} \tag{10}$$

In the formula: K_{as} refers to the power distribution information acquisition rate, N_{as} refers to the number of distribution transformers for information collection, N_{nm} refers to the total number of distribution transformers.

- Efficiency in Power Distribution Automation Coverage: This denotes the proportion of 10kV lines effectively covered by distribution automation within the region, in relation to the total number of 10kV lines in the region.

$$K_c = \frac{N_{dc}}{N_{10a}} \tag{11}$$

In the formula: K_c refers to the effective coverage rate of power distribution, N_{dc} refers to the number of 10kV lines effectively covered by distribution automation, N_{10a} refers to the total number of 10kV lines.

- Rate of Smart-Meter Deployment: This represents the percentage of smart meters installed relative to the overall number of meters installed.

$$K_{sm} = \frac{N_{sm}}{N_{un}} \tag{12}$$

In the formula: K_{sm} refers to the smart-meter coverage rate; N_{sm} refers to the number of smart meter users; N_{um} refers to the total number of electricity users.

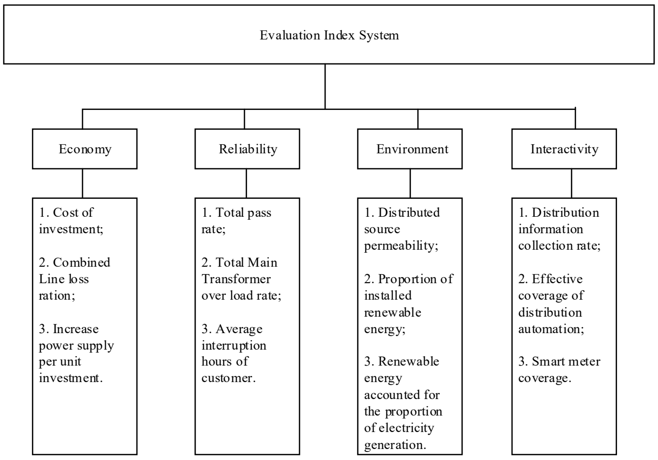

Figure 1. Evaluation index system diagram

B. *Comprehensive evaluation method*

In the comprehensive system evaluation, the system first empowers the index, and subsequently, the weight analysis method is applied to determine the final evaluation result. This process is especially critical for selecting the weighting method and the weight evaluation method. The typical systematic evaluation method mainly analyzes the subjective weighting or objective data. When utilizing any one of them in isolation, the evaluation results may become excessively subjective or overly objective. To strike a balance, a combined subjective and objective weighting method, known as the comprehensive weighting method, can be employed. This approach aims to integrate both subjective and objective considerations for a more balanced and nuanced evaluation.

The Analytic Hierarchy Process (AHP) is frequently employed for assigning subjective weights. Within AHP, the assignment of weights at each level has a direct or indirect impact on the final outcome. It quantifies the influence level of each factor at every stage of the process, enhancing clarity in the evaluation. This method is particularly valuable for systematically assessing complex characteristics and scenarios with multiple objectives, criteria, and time frames. AHP typically starts with the evaluator comprehending the nature and elements of the evaluation problem, giving more priority to qualitative analysis and judgment than conventional quantitative methods.

The Entropy Weight Method is commonly employed as an objective weighting technique. It stands out in capturing the distinctive features of indices, allowing for a more objectively determined set of optimal weights. This method is founded on a theoretical basis, enhancing the credibility of weight assignment. Its application in objective weighting enhances the evaluation process by highlighting the intrinsic characteristics and contributions of each index.

Hence, the index weight employs a blend of subjective and objective weighting. The subjective weight is determined using the hierarchical analysis method to ascertain its weight, while the objective weight is assessed through the entropy weight method. This approach aims to minimize subjective errors and enhance the scientific and rational nature of the results.

1) *Basic steps of analytic hierarchy process*

- Develop a hierarchical structure model by creating a distinct hierarchy aligned with the evaluation objectives and shaping an evaluation index system.
- Constructing the judgment matrix involves forming a comparative judgment matrix for each pair of indices layer by layer. The judgment matrix, denoted as X, is expressed as follows:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \cdots & \cdots & \cdots & \cdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \tag{13}$$

Where x_{ij} indicates the importance of x_i relative to x_j.

- Perform a consistency check, the formula for calculating the consistency index CR is as follows:

$$CR = \frac{CI}{RI} \tag{14}$$

In the equation, if CR is below 0.1, the consistency requirement for the judgment matrix is satisfied. CI represents the consistency index, while RI is the average random consistency index. The computation formula for CI is as follows:

$$CI = \frac{\lambda_{\max} - n}{n-1} \tag{15}$$

- Following the consistency assessment, the vector of $\lambda_{\max}$ the maximum eigenvalue of A is determined, serving as the normalized weight of the index.

2) *Basic steps of the entropy weight method*

- Select the original data matrix, the calculation formula is as follows:

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix} \tag{16}$$

In the formula $r_{ij(m*n)}$ represents the jth evaluation value of the ith evaluation object.

- Compute the characteristic ratio of the ith assessment object concerning the jth index using the following formula:

$$p_{ij} = r_{ij} / \sum_{i=1}^{n} r_{ij} \tag{17}$$

- Calculate the entropy of the jth index, the calculation formula is as follows:

$$e_j = -k \sum_{n=1}^{n} p_{ij} Inp_{ij} \tag{18}$$

- Calculate the difference coefficient of the jth index, the calculation formula is as follows:

$$g_j = 1 - e_j \tag{19}$$

- Establish the weight coefficient for the jth index using the following calculation formula:

$$w_j = e_j / \sum_{j=1}^{m} e_j \tag{20}$$

3) Comprehensive weighting method

The method of subjective weighting involves experts assigning weights based on their personal experience and subjective judgment of the actual situation. In this approach, experts directly provide weights for indices, and the rationality of these weights is influenced by the subjective knowledge of the experts. On the flip side, the comprehensive weighting method merges subjective and objective weighting approaches. This strategy aims to harmonize the judgment of significant indices by experts with the importance of the inherent information in the data. By merging both subjective expert input and objective criteria, the integrated weighting method seeks to generate more rational and well-balanced weights for the evaluation indices.

If the weights derived subjectively using the Analytic Hierarchy Process (AHP) are:

$$z = \begin{bmatrix} z_1 & z_2 & \cdots & z_n \end{bmatrix} \tag{21}$$

If the weights determined objectively using the Entropy Weight Method are:

$$w = \begin{bmatrix} w_1 & w_2 & \cdots & w_n \end{bmatrix} \tag{22}$$

Then the calculation formula of the combined empowerment is:

$$u = \beta z + (1\text{-}\beta) w \tag{23}$$

In Formula (23) : β —Resolution coefficient usually take 0.5.

C. *Equations*

To acquire the ultimate comprehensive assessment outcomes, the comprehensive evaluation model is formulated in the subsequent manner:

$$P = \sum S_{ij} F_{ij} \tag{24}$$

In the formula: S_{ij} represents the weight of the jth index under the ith index level, F_{ij} represents the score value of the jth index below the ith index level.

Using the previously established index system for the thorough evaluation of intelligent distribution networks, a two-tiered system covering economy, reliability, environmental protection, and interactivity is devised. The depiction of the comprehensive evaluation model for the strategic development of intelligent distribution networks is provided below:

$$P = S_1 \times F_1 + S_2 \times F_2 + S_3 \times F_3 + S_4 \times F_4 \tag{25}$$

In the formula: S_1, S_2, S_3, S_4 respectively represents the weight of the first-level indices of economy, reliability, environmental protection and interactivity. And $S_1 + S_2 + S_3 + S_4 = 1$; F_1, F_2, F_3, F_4 individually denote the scores corresponding to the primary indices of economy, reliability, environmental protection, and interactivity. The calculation formula is as follows:

$$F_i = \sum s_{ij} f_{ij} \quad (\iota\text{=1,2,3,4,5}) \tag{26}$$

In the formula: s_{ij} represents the weight of the second-level indices under the first-level indices, f_{ij} represents the evaluated value of the second-tier indices beneath the first-tier indices.

III. DETERMINATION OF INDEX WEIGHT AND SELECTION OF JUDGING BASIS

For primary indices: Using formulas (13) to (15) as mentioned earlier, the weights assigned to economy, reliability, environmental protection, and interactivity for primary indices are as follows: $z = \begin{bmatrix} 0.33 & 0.47 & 0.13 & 0.07 \end{bmatrix}$.

Likewise, the weights of the second-tier indices beneath the primary index economy—investment cost, comprehensive line loss rate, and increased power supply per unit investment—are as follows: $z = \begin{bmatrix} 0.54 & 0.30 & 0.16 \end{bmatrix}$.

The weights of the total pass rate of the secondary index, the overall overload rate of transformers, and the average outage time of users under the reliability of the primary index are as follows: $z = \begin{bmatrix} 0.24 & 0.14 & 0.62 \end{bmatrix}$.

The weights assigned to the second-tier indices within the primary index environmental protection—distributed power supply penetration rate, ratio of installed renewable energy, and proportion of power generation from renewable energy—are as follows: $z = \begin{bmatrix} 0.5 & 0.25 & 0.25 \end{bmatrix}$.

That is, through the use of analytic hierarchy process to calculate 12 indices. The initial weight is:0.1782, 0.099, 0.0528, 0.1128, 0.0658, 0.2914, 0.065, 0.0325, 0.0325, 0.065, 0.0325, 0.0325.

According to equation (18), the entropy values for the indices (A) are as follows: 0.48, 0.622, 0.441, 0.514, 0.5, 0.562, 0.434, 0.631, 0.617, 0.631, 0.419, 0.631.

By using entropy e_j the index difference coefficient g_j is obtained according to equation (19):0.52, 0.378, 0.559, 0.486, 0.5, 0.438, 0.566, 0.369, 0.383, 0.369, 0.581, 0.369.

Then, the weight coefficients w_j of 12 indices can be obtained from equation (20) as:0.094, 0.068, 0.101, 0.088, 0.091, 0.079, 0.103, 0.067, 0.069, 0.067, 0.105, 0.068.

Utilizing Equation (23), the weight coefficient is applied to adjust the initial weights derived from the Analytic Hierarchy Process, resulting in the ultimate calculation of the combined weights for the 12 indices as follows: 0.1361, 0.0835, 0.0769, 0.1004, 0.0784, 0.1852, 0.084, 0.04975, 0.05075, 0.051, 0.06125, 0.04275.

TABLE 1 Evaluation and judgment basis of comprehensive evaluation indices

First-level indices	**Second-level indices**	**Evaluation and judgment basis of indices**					
		0	20	40	60	80	100
Economy	Investment cost (100 million yuan)	30	26	22	18	14	10
	Comprehensive line loss rate (%)	4.2	4.0	3.8	3.6	3.4	3.2
	Increased power supply per unit investment (kWh/ ten thousand yuan)	0	4000	8000	12000	16000	20000
Reliability	Total pass rate (%)	0	20	40	60	80	100
	The overall overload rate of transformers (%)	10	8	6	4	2	0
	Average power outage time of users (h)	20	16	12	8	4	0
Environmental protection	Distributed power supply penetration rate (%)	0	2	4	6	8	10
	The installed capacity ratio of renewable energy (%)	0	20	40	60	80	100
	The ratio of renewable energy generation to electricity generation (%)	0	20	40	60	80	100
Interactivity	Power distribution information acquisition rate (%)	95	96	97	98	99	100
	Effective coverage rate of power distribution automation (%)	0	20	40	60	80	100
	Smart-meter coverage rate (%)	0	20	40	60	80	100

When the index values fall within a specified range, the distribution network can guarantee effective operation. By integrating the values of certain indices provided in the State Grid's guide for the planning and design of distribution networks, combining the insights of various experts in the field, the evaluation standards for the indices are definitively determined. The precise standards are outlined in the table 1.

IV. Example Analysis

This study utilizes the urban distribution network planning data from the electric power company of a city in 2021 as an illustrative case. The comprehensive evaluation system and methodology devised in this study are utilized to comprehensively evaluate the urban distribution network in the city. This process aims to validate the rationality and practicality of the research presented in this paper.

The functionality of the distribution network in this city is explained considering distribution network economy, reliability, environmental protection, and interactivity. Based on the previously defined indices, a comprehensive scoring of the power grid in this city region for the year 2021 is conducted, and the comprehensive score of the power distribution network in city in 2021 are outlined in the table 2. The specific procedures are outlined as follows.

1) *Economic analysis*

- Investment Cost: In 2021, the investment cost of the city's power grid amounted to 1996.04 million yuan, resulting in a calculated score of 6.83 points based on the assigned weight (out of a maximum score of 13.61 points);
- Overall Line Loss Rate: In 2021, the power grid in the city experienced a comprehensive line loss rate of 3.6%, yielding a score of 5.01 points when calculated based on the assigned weight (out of a maximum score of 8.35 points);
- Enhanced power delivery efficiency per unit investment: In the year 2021, the city's power grid achieved a rate of 10100 (kWh / ten thousand yuan) for increased power supply per unit investment, resulting in a score of 3.88 points when assessed based on the assigned weight (with a full score of 7.69 points);

2) *Reliability analysis*

- Total Acceptance Rate: In 2021, the city's power grid achieved an overall acceptance rate of 59.8%, resulting in a score of 6.00 points when assessed based on the assigned weight (out of a maximum score of 10.04 points);
- The overall overload rate of transformers: in 2021, the overall overload rate of transformers of city power grid

was 4.4%, and the score calculated according to the weight was 4.39 points (The full score was 7.84 points);

- Mean downtime for electricity users: In 2021, the average duration of power interruptions for users in the city's power grid was 9.02 hours, yielding a score of 10.17 points based on the assigned weight (with a maximum score of 18.52 points);

3) Environmental protection analysis

- Penetration Rate of Decentralized Power Supply: In 2021, the distributed power supply penetration rate within the city's power grid reached 3.55%, resulting in a score of 2.98 points when evaluated based on the assigned weight (out of a maximum score of 8.4 points);
- Ratio of Renewable Energy Installed Capacity: In 2021, the city's power grid achieved an 80% installed capacity ratio for renewable energy, earning a score of 3.98 points (out of a possible 4.975 points) based on the assigned weight;
- Renewable Energy's Share in Total Electricity Generation: In 2021, the power grid in the city accomplished an 80% proportion of renewable energy generation to overall electricity generation. resulting in a score of 4.06 points when evaluated based on the assigned weight (with a full score of 5.075 points);

4) Interaction analysis

- Rate of Power Distribution Information Retrieval: In 2021, the city power grid achieved a 100% rate in acquiring power distribution information, resulting in a score of 5.1 points when assessed based on the assigned weight (with a full score of 5.1 points);
- Efficiency in Power Distribution Automation Coverage: In 2021, the city power grid achieved a 39.35% effective coverage rate of power distribution automation, resulting in a score of 2.41 points when evaluated based on the assigned weight (with a full score of 6.125 points);
- Intelligent Meter Adoption Rate: In 2021, the power grid in the city attained a 100% adoption rate of smart meters, resulting in a score of 4.275 points when assessed based on the assigned weight (with a full score of 4.275 points).

TABLE 2 Comprehensive score of the power distribution network in city in 2021

Year	Category	Total score	First-level indices score			
			Economy	Reliability	Environmental protection	Interactivity
In 2021	Full mark	100	29.65	36.4	18.45	15.5
	Actual score	59.085	15.72	20.56	11.02	11.785
	Lost score	40.915	13.93	15.84	7.43	3.715

V. CONCLUSION

In this study, we introduce a holistic weighting approach using the AHP-entropy weight method for the prospective intelligent integrated energy system in distribution networks. Based on the simulation example, the findings are as follows:

- After applying the evaluation system in this urban region, the comprehensive assessment results clearly reflected the overall development stage of the upcoming distribution network. They generally aligned with the economic and social positioning of the city, thus affirming the applicability and robustness of the comprehensive evaluation system and approach proposed in this research.
- Regarding the economic aspect, the power grid investment in 2021 was excessively elevated, leading to a diminished value of increased power supply per unit investment. Due to influences from the management system, the comprehensive line loss rate was comparatively elevated, consequently contributing to a reduced economic score.
- In terms of reliability, there was no reasonable diversion of heavy load lines and balanced line load rate, there were invalid connections and high line load rate, resulting in a low total pass rate. Furthermore, the users experienced prolonged average power outage durations, leading to a diminished reliability score.
- In terms of environmental protection, the main factor affecting its score was that there were fewer distributed power supplies, resulting in low permeability of distributed power supply penetration rate.
- In terms of interactivity, the main factor affecting its score was the low distribution automation efficiency coverage of 10kV lines.

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